



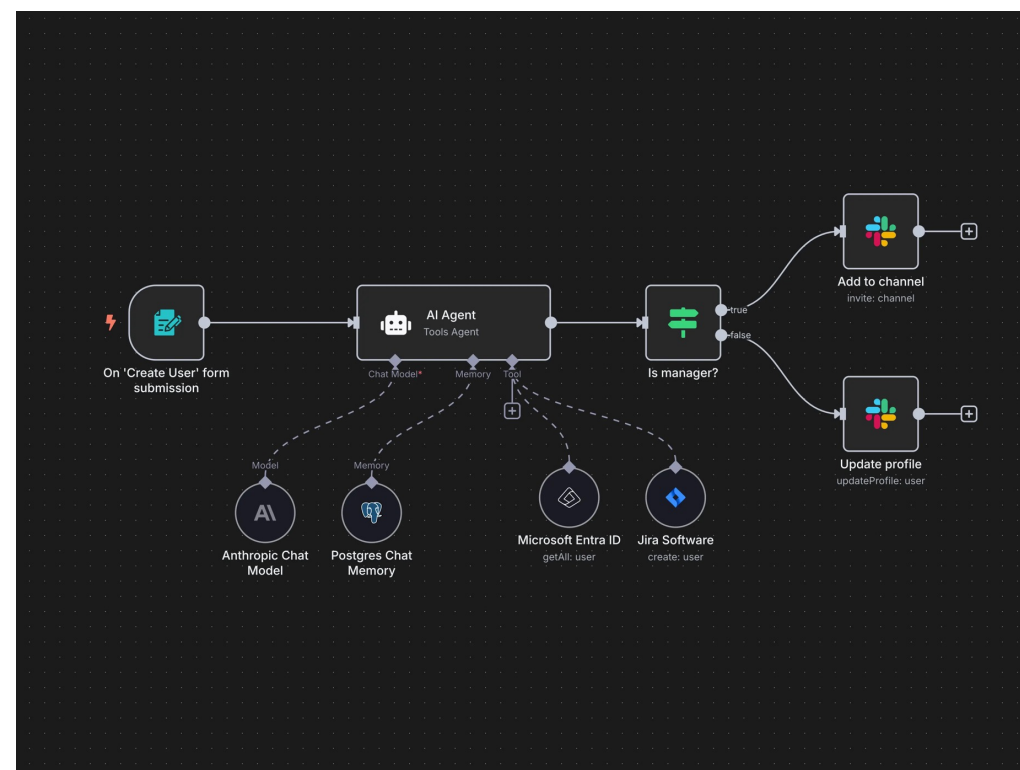
N8N

Prof. Peerapon Vateekul, Ph.D.

Peerapon.v@chula.ac.th

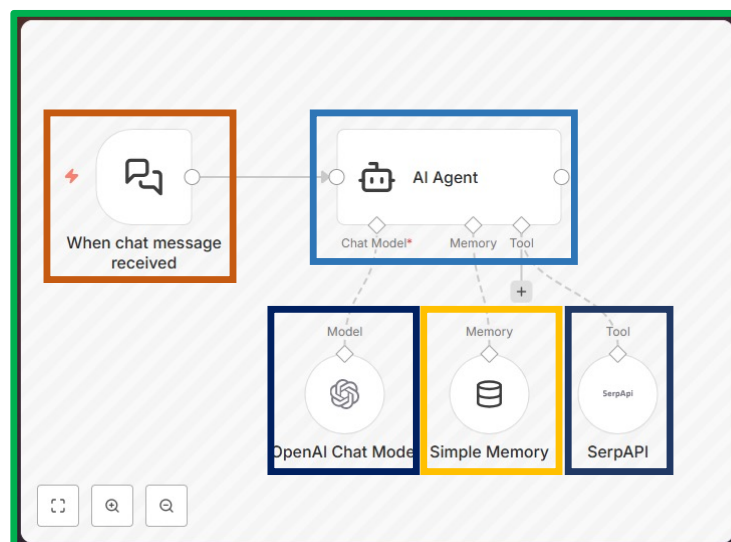
What is N8N?

n8n (pronounced “n-eight-n”) is a workflow automation platform that helps you connect apps, move data, and automate tasks without needing to write a lot of code. You can think of it as a visual tool for automating repetitive work by designing flows that link different services together.

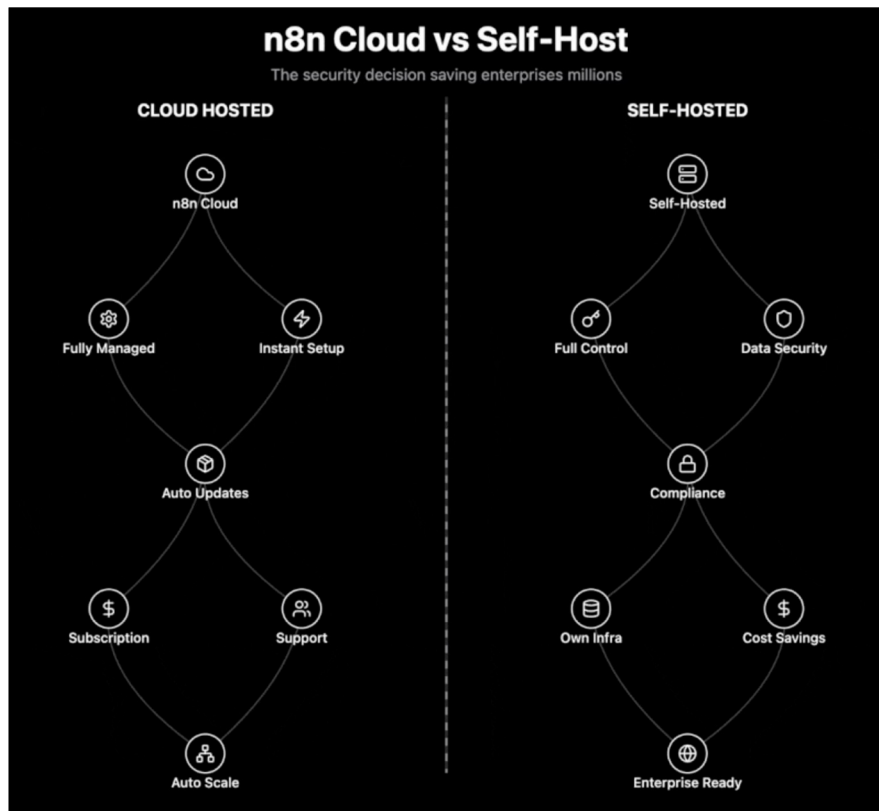


N8N Components

1. **Workflow:** The complete automation that connects all nodes to handle a chat message from start to finish.
2. **Node:** A single functional step in the workflow that performs a specific action. In n8n there are a lot of node types. for example,
 1. **Trigger Node:** The entry point of the workflow that starts execution when get notified (such as when receive messages).
 2. **AI Agent Node:** The main orchestration node that processes user input and orchestrates models, memory, and tools.
 3. **Memory Node:** A sub-node attached to the AI Agent that stores and retrieves conversational context.
 4. **Chat Model Node:** LLM/Chat model node that will receive input and get generative response.
3. **Tools:** External capabilities (e.g., search or APIs) that the AI Agent can call to enrich its response.



Where to use?



1) Localhost (your computer) - setup require Docker/Node.js

2) Self-hosted (your own server) - you are responsible for **deployment**, security, updates, and uptime

✓ 3) n8n Cloud (managed service) — used in this lab

- n8n hosts and maintains the platform for you (no server setup)
- Best for: fastest start, reliable uptime, easy sharing/collaboration
- **In this lab:** use the **14-day free trial** to build workflows without installation

Start automating today

Full name

Company email

Confirm email address

Password

Account name

.app.n8n.cloud☐

Keep up with our constant improvements through our newsletter



Success!



Start free 14-day trial

By continuing, you agree to our [terms and conditions](#),
[Privacy policy](#).

Registration & Free Trial Activation (n8n Cloud)

Go [here](#).

Let's Drag, Drop, and Automate

What we will build in n8n (using the built-in Chat UI)

Lab 8.0 – Simple Q&A Chatbot

Lab 8.1 – Medical Research Assistant with PubMed

- Input: a natural-language question in chat
- Workflow: validate the request → create a structured PubMed search plan → call PubMed APIs → format a concise answer with citations (PMID/title/journal/year) → respond in chat.

Lab 8.2 – Document-based RAG Chatbot

- Input: either (a) upload a document, or (b) ask a question
 - Workflow:
 - Upload path: load document → chunk text → create embeddings → store in a vector database
 - Question path: retrieve relevant chunks → generate an answer grounded in retrieved text → respond in chat.
-

+ Thank you
& any questions